

This agrees with the original definition of cup product since we have $\Delta^*(a \times b) = \Delta^*(p_1^*(a) \smile p_2^*(b)) = \Delta^*(p_1^*(a)) \smile \Delta^*(p_2^*(b)) = a \smile b$, as both compositions $p_1\Delta$ and $p_2\Delta$ are the identity map of X .

Unfortunately, the definition of cellular cross product cannot be combined with Δ to give a definition of cup product at the level of cellular cochains. This is because Δ is not a cellular map, so it does not induce a map of cellular cochains. It is possible to homotope Δ to a cellular map by Theorem 4.8, but this involves arbitrary choices. For example, the diagonal of a square can be pushed across either adjacent triangle. In particular cases one might hope to understand the geometry well enough to compute an explicit cellular approximation to the diagonal map, but usually other techniques for computing cup products are preferable.

The cohomology cross product satisfies the same commutativity relation as for homology, namely $T^*(a \times b) = (-1)^{k\ell} b \times a$ for $T: X \times Y \rightarrow Y \times X$ the transposition map, $a \in H^k(Y; R)$, and $b \in H^\ell(X; R)$. The proof is the same as for homology. Taking $X = Y$ and noting that $T\Delta = \Delta$, we obtain a new proof of the commutativity property of cup product.

Exercises

1. Compute the groups $H_i(\mathbb{RP}^m \times \mathbb{RP}^n; G)$ and $H^i(\mathbb{RP}^m \times \mathbb{RP}^n; G)$ for $G = \mathbb{Z}$ and $\mathbb{Z}_2$ via the cellular chain and cochain complexes. [See Example 3B.4.]
2. Let C and C' be chain complexes, and let I be the chain complex consisting of $\mathbb{Z}$ in dimension 1 and $\mathbb{Z} \times \mathbb{Z}$ in dimension 0, with the boundary map taking a generator e in dimension 1 to the difference $v_1 - v_0$ of generators v_i of the two $\mathbb{Z}$'s in dimension 0. Show that a chain map $f: I \otimes C \rightarrow C'$ is precisely the same as a chain homotopy between the two chain maps $f_i: C \rightarrow C'$, $c \mapsto f(v_i \otimes c)$, $i = 0, 1$. [The chain homotopy is $h(c) = f(e \otimes c)$.]
3. Show that the splitting in the topological Künneth formula cannot be natural by considering the map $f \times \mathbb{1}: M(\mathbb{Z}_m, n) \times M(\mathbb{Z}_m, n) \rightarrow S^{n+1} \times M(\mathbb{Z}_m, n)$ where f collapses the n -skeleton of $M(\mathbb{Z}_m, n) = S^n \cup e^{n+1}$ to a point.
4. Show that the cross product of fundamental classes for closed R -orientable manifolds M and N is a fundamental class for $M \times N$.
5. Show that **slant products**

$$H_n(X \times Y; R) \times H^j(Y; R) \rightarrow H_{n-j}(X; R), \quad (e^i \times e^j, \varphi) \mapsto \varphi(e^j) e^i$$

$$H^n(X \times Y; R) \times H_j(Y; R) \rightarrow H^{n-j}(X; R), \quad (\varphi, e^j) \mapsto (e^i \mapsto \varphi(e^i \times e^j))$$

can be defined via the indicated cellular formulas. [These 'products' are in some ways more like division than multiplication, and this is reflected in the common notation a/b for them, or $a \setminus b$ when the order of the factors is reversed. The first of the two slant products is related to cap product in the same way that the cohomology cross product is related to cup product.]

3.C H-Spaces and Hopf Algebras

Of the three axioms for a group, it would seem that the least subtle is the existence of an identity element. However, we shall see in this section that when topology is added to the picture, the identity axiom becomes much more potent. To give a name to the objects we will be considering, define a space X to be an **H-space**, ‘H’ standing for ‘Hopf,’ if there is a continuous multiplication map $\mu: X \times X \rightarrow X$ and an ‘identity’ element $e \in X$ such that the two maps $X \rightarrow X$ given by $x \mapsto \mu(x, e)$ and $x \mapsto \mu(e, x)$ are homotopic to the identity through maps $(X, e) \rightarrow (X, e)$. In particular, this implies that $\mu(e, e) = e$.

In terms of generality, this definition represents something of a middle ground. One could weaken the definition by dropping the condition that the homotopies preserve the basepoint e , or one could strengthen it by requiring that e be a strict identity, without any homotopies. An exercise at the end of the section is to show the three possible definitions are equivalent if X is a CW complex. An advantage of allowing homotopies in the definition is that a space homotopy equivalent in the basepointed sense to an H-space is again an H-space. Imposing basepoint conditions is fairly standard in homotopy theory, and is usually not a serious restriction.

The most classical examples of H-spaces are **topological groups**, spaces X with a group structure such that both the multiplication map $X \times X \rightarrow X$ and the inversion map $X \rightarrow X$, $x \mapsto x^{-1}$, are continuous. For example, the group $GL_n(\mathbb{R})$ of invertible $n \times n$ matrices with real entries is a topological group when topologized as a subspace of the n^2 -dimensional vector space $M_n(\mathbb{R})$ of all $n \times n$ matrices over $\mathbb{R}$. It is an open subspace since the invertible matrices are those with nonzero determinant, and the determinant function $M_n(\mathbb{R}) \rightarrow \mathbb{R}$ is continuous. Matrix multiplication is certainly continuous, being defined by simple algebraic formulas, and it is not hard to see that matrix inversion is also continuous if one thinks for example of the classical adjoint formula for the inverse matrix.

Likewise $GL_n(\mathbb{C})$ is a topological group, as is the quaternionic analog $GL_n(\mathbb{H})$, though in the latter case one needs a somewhat different justification since determinants of quaternionic matrices do not have the good properties one would like. Since these groups GL_n over $\mathbb{R}$, $\mathbb{C}$, and $\mathbb{H}$ are open subsets of Euclidean spaces, they are examples of *Lie groups*, which can be defined as topological groups which are also manifolds. The GL_n groups are noncompact, being open subsets of Euclidean spaces, but they have the homotopy types of compact Lie groups $O(n)$, $U(n)$, and $Sp(n)$. This is explained in §3.D for $GL_n(\mathbb{R})$, and the other two cases are similar.

Among the simplest H-spaces from a topological viewpoint are the unit spheres S^1 in $\mathbb{C}$, S^3 in the quaternions $\mathbb{H}$, and S^7 in the octonions $\mathbb{O}$. These are H-spaces since the multiplications in these division algebras are continuous, being defined by